

On polarities and computational convexity

Frank Nielsen

Sony Computer Science Laboratories, Inc.

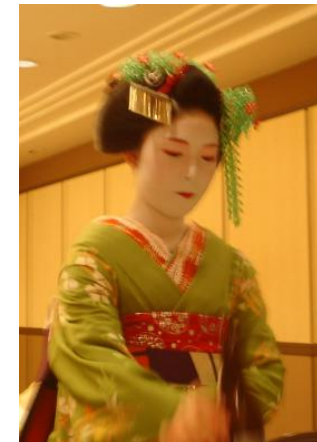
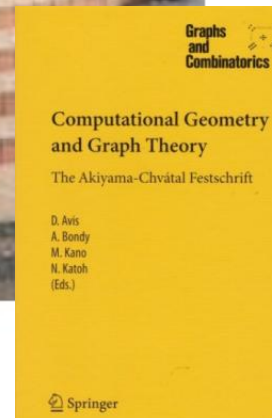


10th September 2026

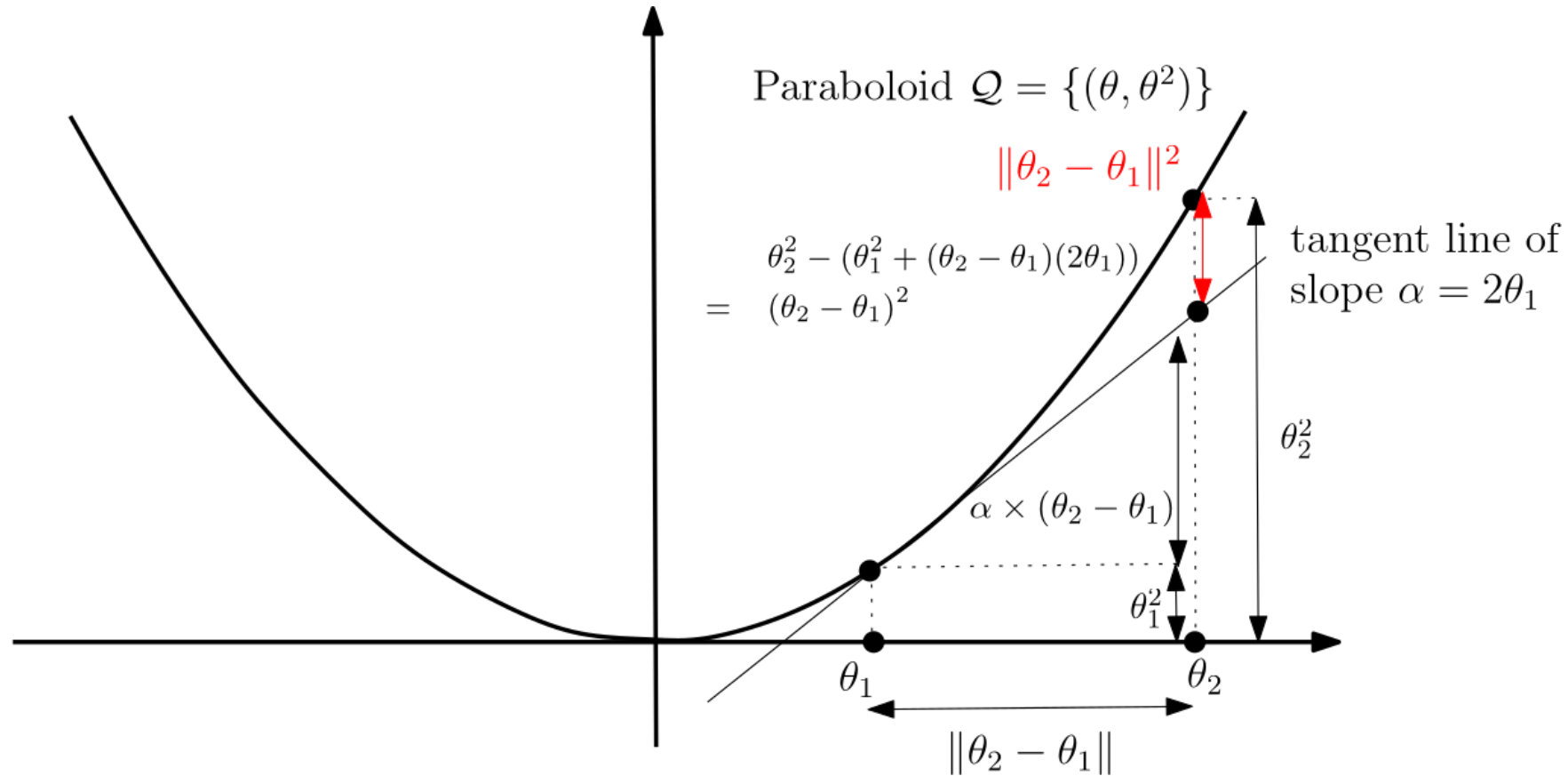
Background: G (1998) $\rightarrow$ GG (2007) $\rightarrow$ GGG (2026)



Fig. 14. Participants at JCDCG 1998



A view of the squared Euclidean distance



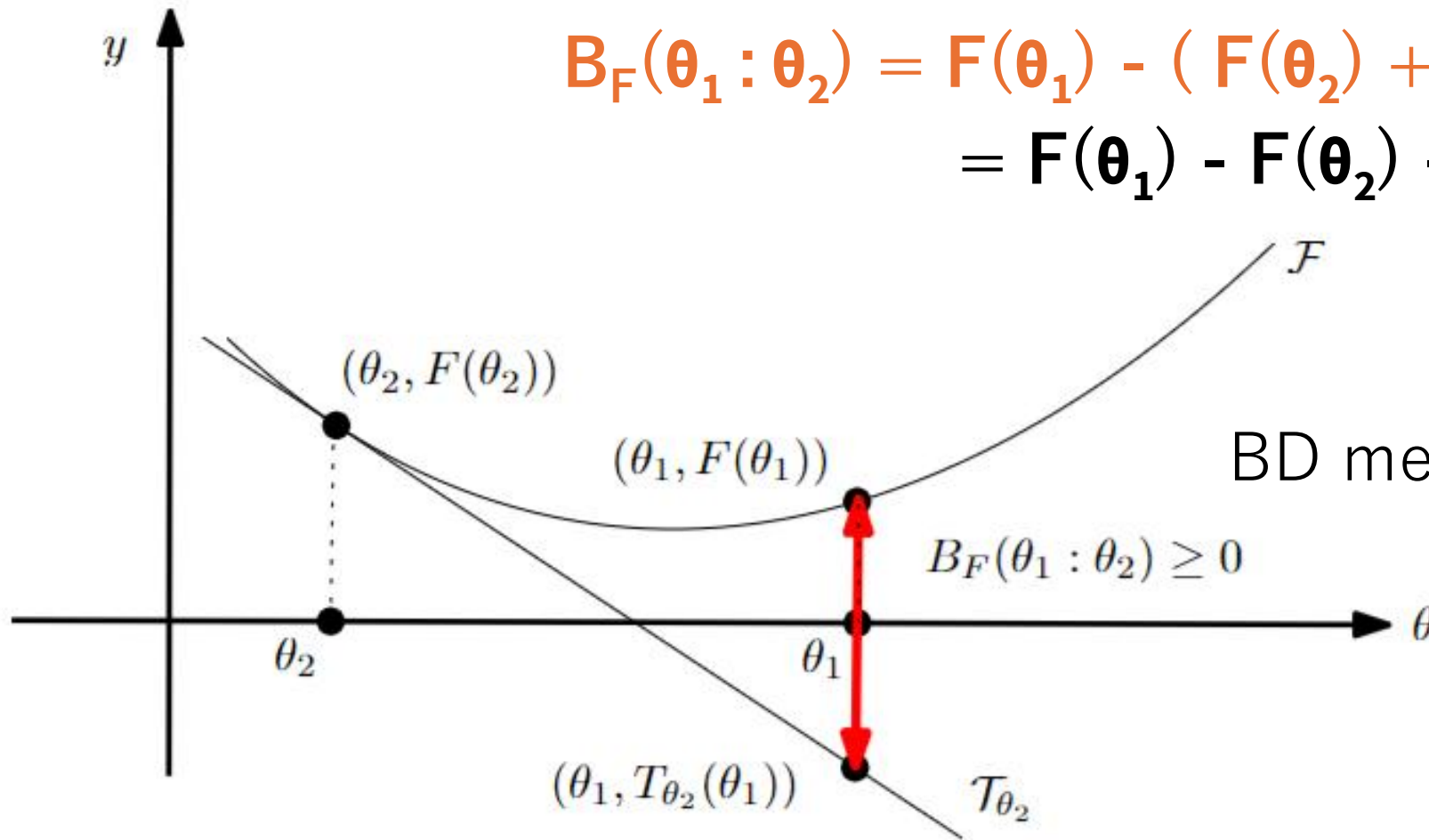
- Squared Euclidean distance is *not a metric* because it violates the triangle inequality.
 - Smooth distances are called **divergences**
- (Related to the *paraboloid lifting* in Computational Geometry for Voronoi/Delaunay)

Bregman divergences: Convex potential functions

- $F: \Theta \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ a strictly convex and smooth real-valued function on a finite-dimensional Hilbert space equipped with inner product $\langle \cdot, \cdot \rangle$

Bregman divergence $B_F: \Theta \times \text{Int}(\Theta) \rightarrow \mathbb{R}$

$$\begin{aligned} B_F(\theta_1 : \theta_2) &= F(\theta_1) - (F(\theta_2) + \langle \theta_1 - \theta_2, \nabla F(\theta_2) \rangle) \\ &= F(\theta_1) - F(\theta_2) - \langle \theta_1 - \theta_2, \nabla F(\theta_2) \rangle \end{aligned}$$



Bregman divergences unify important distortions

$$B_F(\theta_1 : \theta_2) = F(\theta_1) - F(\theta_2) - \langle \theta_1 - \theta_2, \nabla F(\theta_2) \rangle$$

- **Quadratic negentropy** generator yields ***squared Euclidean divergence***:

$$F(\theta) = \frac{1}{2} \theta^T Q \theta \text{ with } \Theta = \mathbb{R}^d$$

Computational geometry L_2^2

$$\nabla F(\theta) = Q \theta$$

$$B_F(\theta_1 : \theta_2) = \frac{1}{2} (\theta_2 - \theta_1)^T Q (\theta_2 - \theta_1) = \frac{1}{2} \|\theta_2 - \theta_1\|_Q^2$$

- **Shannon negentropy** generator yields ***relative entropy*** as known as *Kullback-Leibler divergence*:

$$F(\theta) = \sum_i \theta_i \log(\theta_i) \text{ with } \Theta = \Delta^d \text{ (standard/probability simplex)}$$

$$\nabla F(\theta) = [\log(\theta_i) - \theta_i]_i$$

$$B_F(\theta : \theta') = \sum_i \{ \theta_i \log(\theta_i / \theta'_i) + \theta'_i - \theta_i \} = \sum_i \theta_i \log(\theta_i / \theta'_i)$$

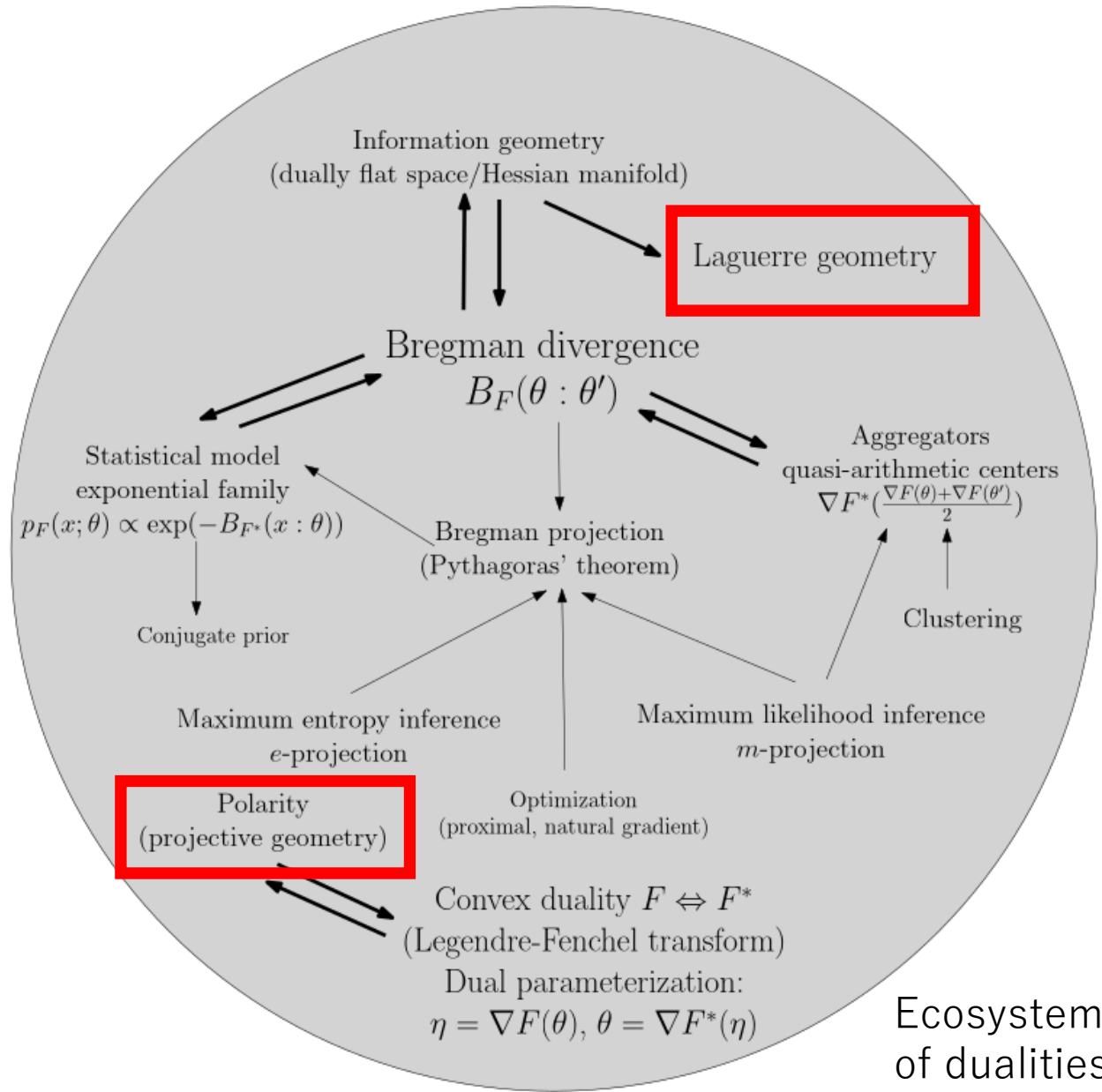
$$= \text{KL}(\theta_i : \theta') \text{ since } \sum_i \theta_i = \sum_i \theta'_i = 1$$

Information geometry

Bregman divergences are the workhorses in ML and AI

This talk describes two connections of Bregman divergences with computational geometry:

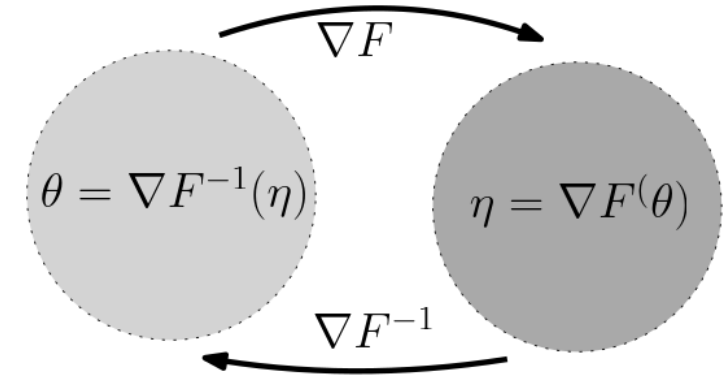
1. Power distances
2. Quadratic Polarities



Dual Bregman/Fenchel-Young divergences

- **Gradient map** ∇F yields **dual parameterization**

$$\eta = \nabla F(\theta)$$



- Build the **Fenchel-Young divergence** from Young inequality:

$$F(\theta_1) + F^*(\eta_2) \geq \langle \theta_1, \eta_2 \rangle$$

$$Y_{F, F^*}(\theta_1, \eta_2) = F(\theta_1) + F^*(\eta_2) - \langle \theta_1, \eta_2 \rangle \geq 0$$

$$B_F(\theta_1 : \theta_2) = Y_{F, F^*}(\theta_1 : \eta_2) = Y_{F^*, F}(\eta_2, \theta_1) = B_{F^*}(\eta_2 : \eta_1)$$

where F^* is the Legendre-Fenchel transform of F :

$$F^*(\eta) = \sup_{\theta \in \Theta} \{ \langle \theta, \eta \rangle - F(\theta) \}$$

$$\text{with } \theta = \nabla F^{-1}(\eta) = \nabla F^*(\eta)$$

Bregman divergences as power distances

BD from Fenchel-Young divergence $Y_F(\theta : \eta') := F(\theta) + F^*(\eta') - \langle \theta, \eta' \rangle$

Then use identity $-\langle \theta, \eta' \rangle = \frac{1}{2} (\|\theta - \eta'\|^2 - \|\theta\|^2 - \|\eta'\|^2)$

and get BD as PD

$$B_F(\theta : \theta') = Y_F(\theta : \eta') = \frac{1}{2} \sigma(\hat{\theta} : \check{\eta}')$$

Power distance
(signed measure)

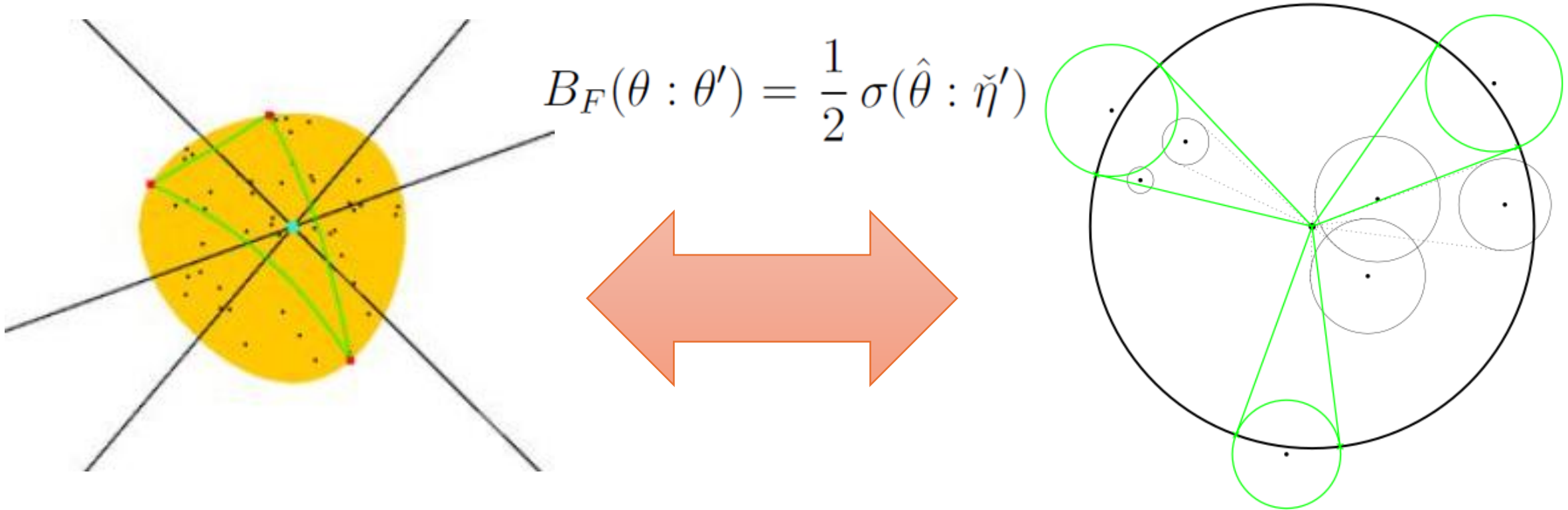
$$\sigma(\hat{p}, \hat{p}') := \|p - p'\|^2 - w - w'$$
$$\hat{p} = (p, w) \quad \hat{p}' = (p', w')$$

Weighted points expressed in **mixed parameterizations**:

$$\hat{\theta} = (\theta, \|\theta\|^2 - 2F(\theta)) \quad \check{\eta} = (\eta, \|\eta\|^2 - 2F^*(\eta))$$

Bregman divergences as power distances: CG

Use toolbox of computational geometry in information geometry!



Bregman miniball in θ

as Power miniball in η

In information theory/ML, **redundancy** measure amounts to BD

ON CONJUGATE CONVEX FUNCTIONS

W. FENCHEL

1. Since the classical work of Minkowski and Jensen it is well known that many of the inequalities used in analysis may be considered as consequences of the convexity of certain functions. In several of these inequalities pairs of “conjugate” functions occur, for instance pairs of powers with exponents a and a related by $1/a + 1/a = 1$. A more general example is the pair of positively homogeneous convex functions defined by Minkowski and known as the distance (or gauge) function and the function of support of a convex body. The purpose of the present paper is to explain the general (by the way rather elementary) idea underlying this correspondence. Subjected to a more

The hypersurfaces $\gamma = f(x_1, \dots, x_n)$ and $\eta = \phi(\xi_1, \dots, \xi_n)$ correspond to each other in the polarity with respect to the paraboloid

$$2y = x_1^2 + \dots + x_n^2.$$

Convex conjugates from quadratic polarities*

*Joint work with Basile Plus-Gourdon and Mahito Sugiyama

Polarity of a **set** A : $\Delta(A) := \{[b] \in \mathbb{R}_{n+1} \quad : \quad \forall [a] \in A, \quad p(a, b) \geq 0\}$

Quadratic polarity when $p(a, b) := [a]^\top C [b]$

Property (Legendre transformation from Legendre polarity). *Let $F : \Theta \rightarrow \mathbb{R}$ be a real-valued function and $F^* : H \rightarrow \mathbb{R}$ be its Legendre convex conjugate. Then it holds that*

[Fenchel 1949]

$$\partial \Delta_{\mathcal{L}}(\text{graph}(F)) = \text{graph}(F^*)$$

Legendre polarity $C = C_{\mathcal{L}} := \begin{bmatrix} -I_n & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

n -variate function $\rightarrow$ Graph $(n+1)$ -dim $\rightarrow$ Projective space/homogeneous coordinates $(n+2)$ -dim

The concept of duality in convex analysis, and the characterization of the Legendre transform

By SHIRI ARTSTEIN-AVIDAN and VITALI MILMAN*

THEOREM 1. *Assume a transform $\mathcal{T} : \text{Cvx}(\mathbb{R}^n) \rightarrow \text{Cvx}(\mathbb{R}^n)$ (defined on the whole domain $\text{Cvx}(\mathbb{R}^n)$) satisfies*

1. $\mathcal{T}\mathcal{T}\phi = \phi$,
2. $\phi \leq \psi$ implies $\mathcal{T}\phi \geq \mathcal{T}\psi$.

Then, $\mathcal{T}$ is essentially the classical Legendre transform; namely there exists a constant $C_0 \in \mathbb{R}$, a vector $v_0 \in \mathbb{R}^n$, and an invertible symmetric linear transformation $B \in \text{GL}_n$ such that

$$(\mathcal{T}\phi)(x) = (\mathcal{L}\phi)(Bx + v_0) + \langle x, v_0 \rangle + C_0.$$

Legendre transform and gauge transformations*

*Joint work with Basile Plus-Gourdon and Mahito Sugiyama

Quadratic Polarity realized as Legendre Polarity → Deformation

Theorem 1 (Quadratic polarity as transformed convex body of the Legendre polarity). *Let $C \in \text{GL}(n+2)$ and suppose there exists $\mathcal{M}_T \in \text{GL}(n+2)$ such that $C = C_{\mathcal{L}} \mathcal{M}_T^{-1}$, where T is an affine deformation defined by its matrix $\mathcal{M}_T = C^{-1} C_{\mathcal{L}}$ as $T : [b] \mapsto \mathcal{M}_T[b]$. Then for any $A \subset \mathbb{R}^{n+1}$, we have $\Delta_C(A) = T(\Delta_{\mathcal{L}}(A))$.*

Quadratic Polarity realized as Deformation → Legendre Polarity

Theorem 2 (Quadratic polarity as Legendre polarity on deformed convex body). *Let $C \in \text{GL}(n+2)$ and suppose there exists $\mathcal{M}_S \in \text{GL}(n+2)$ such that: $C = \mathcal{M}_S^T C_{\mathcal{L}}$, where S is an affine deformation defined by its matrix $\mathcal{M}_S = C_{\mathcal{L}} C^T$ as $S : [a] \mapsto \mathcal{M}_S[a]$. Then for any $A \subset \mathbb{R}^{n+1}$: $\Delta_C(A) = \Delta_{\mathcal{L}}(S(A))$.*

with correspondences: $\mathcal{M}_T = C_{\mathcal{L}} \mathcal{M}_S^{-T} C_{\mathcal{L}} \longleftrightarrow \mathcal{M}_S = C_{\mathcal{L}} \mathcal{M}_T^{-T} C_{\mathcal{L}}$

Gauge freedom with matrix pipeline

Fenchel-Young divergences as inner products (GPU)

F a Legendre-type strictly convex function with convex conjugate F^* , $(F^*)^*=F$
Fenchel-Young divergences are equivalent to Bregman divergences:

$$\begin{aligned} Y_F(\theta_1 : \eta_2) &:= F(\theta_1) + F^*(\eta_2) - \langle \theta_1, \eta_2 \rangle = Y_{F^*}(\eta_2 : \theta_1) \geq 0 \\ &= \left\langle \begin{bmatrix} -\theta_1 \\ 1 \\ F(\theta_1) \end{bmatrix}, \begin{bmatrix} \eta_2 \\ F^*(\eta_2) \\ 1 \end{bmatrix} \right\rangle \end{aligned} \quad \text{Inner product on graph points using homogeneous coordinates}$$

Geometric interpretation: Legendre transform considered as a **polarity**

$$Y_F(\theta_1 : \eta_2) = (\tilde{\theta}_1^\uparrow)^\top C_{\mathcal{L}} \tilde{\eta}_2^\uparrow, \quad C_{\mathcal{L}} := \begin{bmatrix} -I & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \longrightarrow \quad Y_F(\theta_1 : \eta_2) = \left\langle \begin{bmatrix} -\theta_1 \\ 1 \\ F(\theta_1) \end{bmatrix}, \begin{bmatrix} \eta_2 \\ F^*(\eta_2) \\ 1 \end{bmatrix} \right\rangle$$

where $\tilde{\theta}^\uparrow := \begin{bmatrix} \theta \\ F(\theta) \\ 1 \end{bmatrix}$, $\tilde{\eta}^\uparrow := \begin{bmatrix} \eta \\ F^*(\eta) \\ 1 \end{bmatrix}$ are homogeneous vectors of points on the graphs of potential functions

Convex conjugates F^* via neural networks*

*Joint work with Basile Plus-Gourdon

2606.09077

- Conjugate $F^*(\eta) = \langle (\nabla F)^{-1}(\eta), \eta \rangle - F((\nabla F)^{-1}(\eta))$

may not be in closed form when $(\nabla F)^{-1}$ not available or computationally intractable: Often the case in ML and optimal transport.

- Conjugation map:
$$\theta^*(\eta) = \operatorname{argmax}_{\theta \in \Theta} \{ \langle \theta, \eta \rangle - f(\theta) \} = \nabla f^*(\eta)$$

parameterize a *residual network* $\theta_\varphi(\eta) = \eta + \text{MLP}_\varphi(\eta)$, with the multilayer perceptron (MLP)

Motivation: Half quadratic paraboloid yield identity, linear gradient map

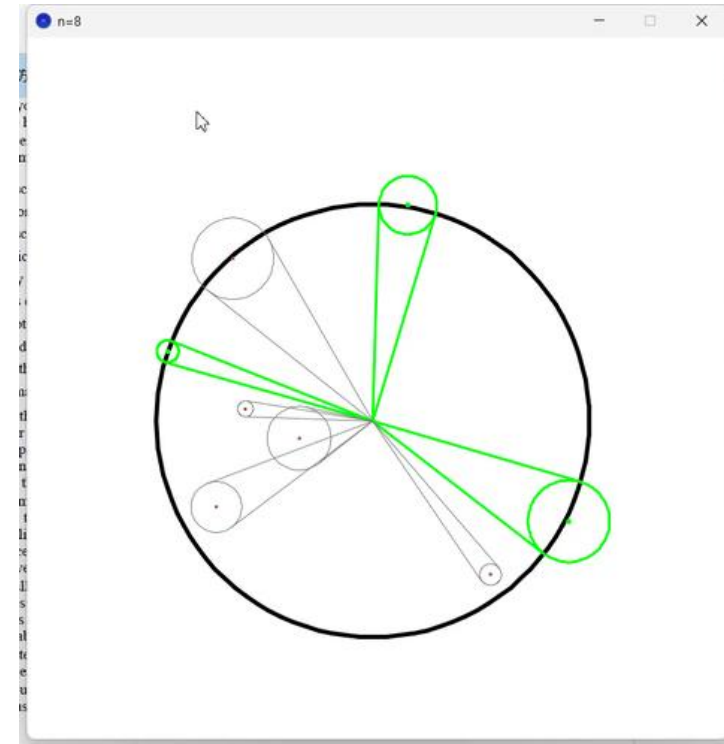
- Train NN with loss function : $\mathcal{L}(\varphi) = \mathbb{E}_\eta [f(\theta_\varphi(\eta)) - \langle \theta_\varphi(\eta), \eta \rangle]$
- Once trained, we *approximate the convex conjugate* by

$$\hat{f}^*(\eta) = \langle \theta_\varphi(\eta), \eta \rangle - f(\theta_\varphi(\eta))$$

- We use **gauge freedom** to **stabilize numerically** the NN learning...

Computational geometry for Machine Learning

- Kullback-Leibler minimax called redundancy in information theory solved as equivalent smallest enclosing power balls
- Computational convexity with NNs



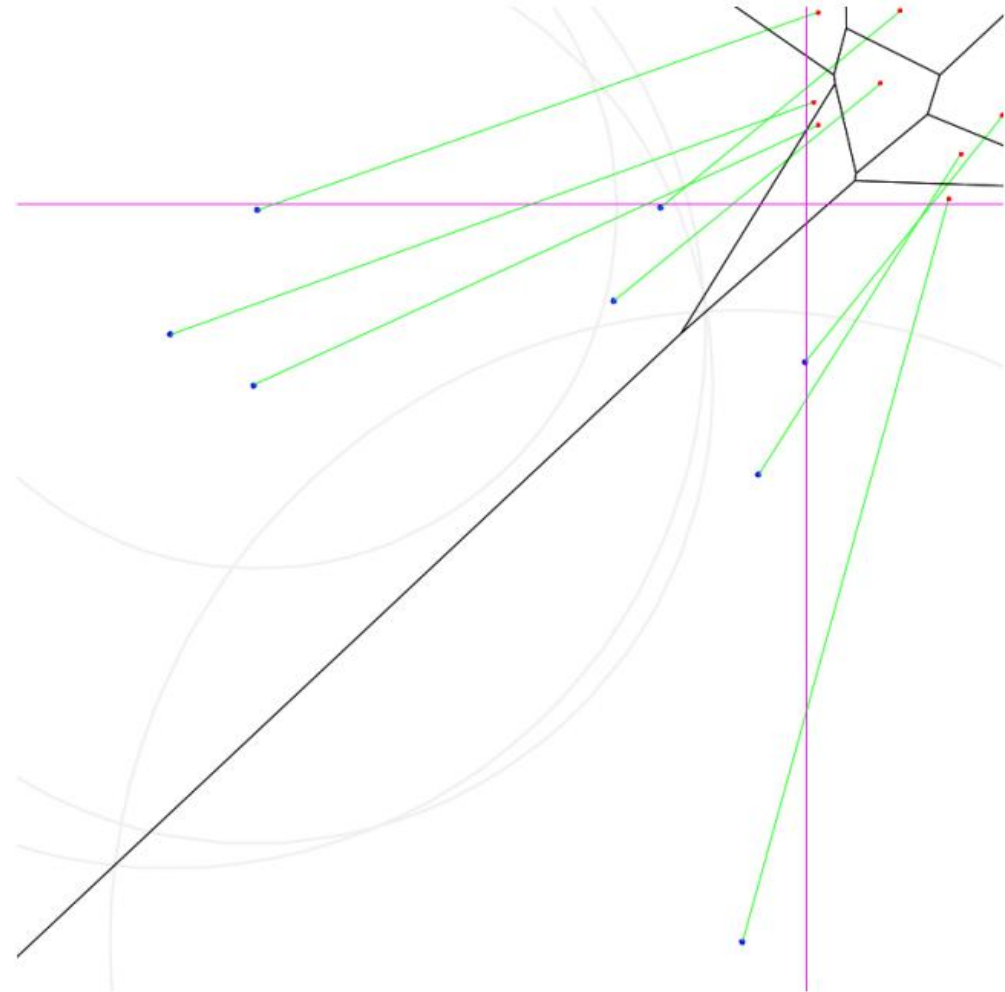
References

- Neural Legendre-Fenchel transform with Hessian Preconditioning, arXiv:2606.09077
- Minimum enclosing Bregman balls made easy, arXiv:2607.24197

Bregman divergences as power distances: CG

Use toolbox of computational geometry in information geometry!

$$B_F(\theta : \theta') = \frac{1}{2} \sigma(\hat{\theta} : \check{\eta}')$$

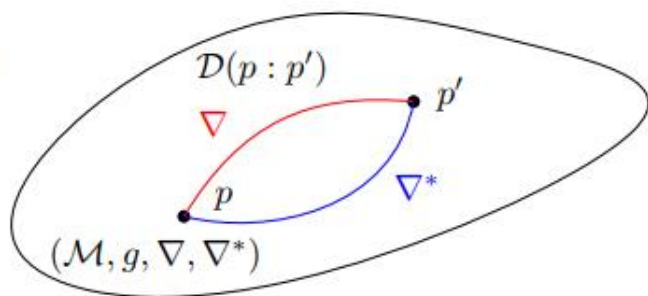


Kullback-Leibler Voronoi diagram

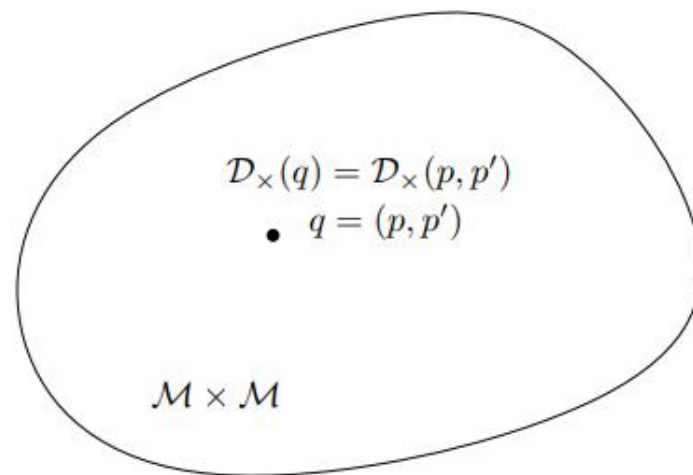
Bregman Voronoi in θ as clipped power diagrams in η

Geometry

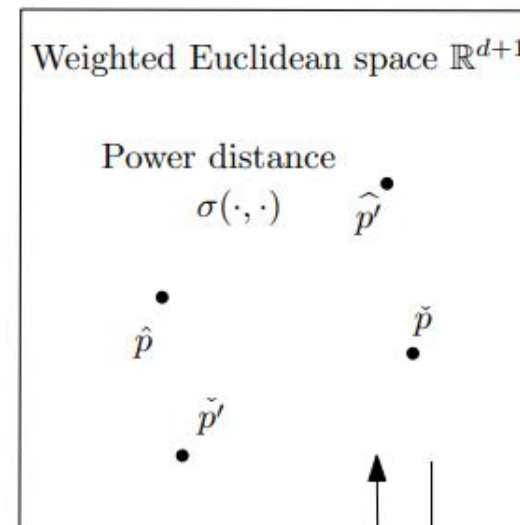
Divergence on Hessian manifold



Contrast function on product manifold

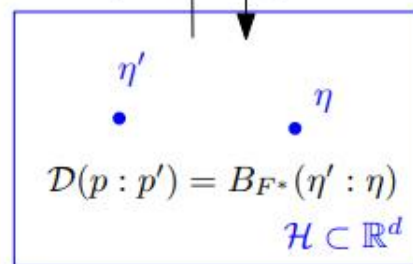
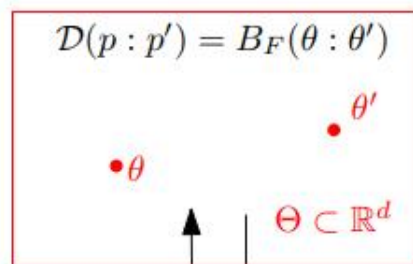


Laguerre geometry



charts

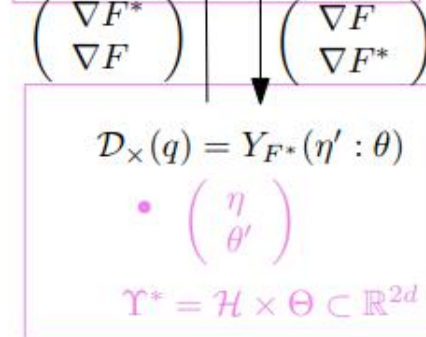
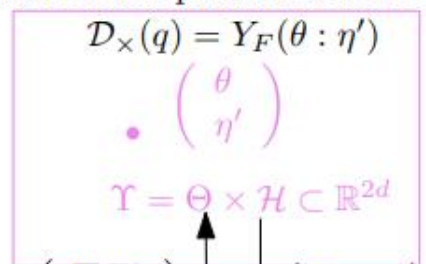
∇ -affine coordinate system



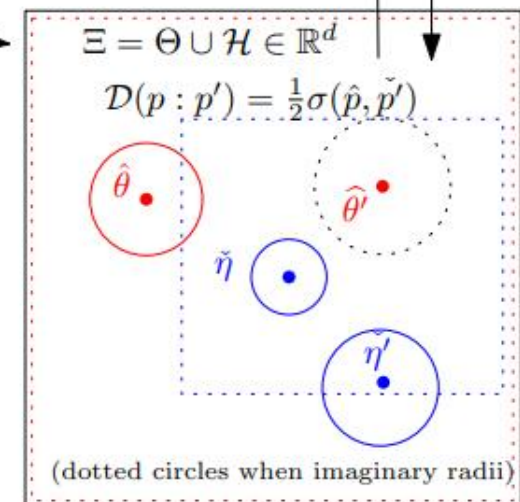
∇^* -affine coordinate system

mapping $\hat{\theta} = (\theta, \|\theta\|^2 - 2F(\theta))$

stacked mixed parameterization



commutator



superposed canonical charts

mapping $\tilde{\eta} = \|\eta\|^2 - 2F^*(\eta)$

A characterization of the duality mapping for convex bodies

Károly J. Böröczky and Rolf Schneider

$$K^* := \{x \in \mathbb{R}^d : \langle x, y \rangle \leq 1 \text{ for all } y \in K\}.$$

We write $A \vee B := \text{conv}(A \cup B)$ for $A, B \subset \mathbb{R}^d$ and $x_1 \vee \cdots \vee x_k := \text{conv}\{x_1, \dots, x_k\}$ for $x_1, \dots, x_k \in \mathbb{R}^d$.

Theorem. *Let $\psi : \mathcal{K}_{(0)}^d \rightarrow \mathcal{K}_{(0)}^d$ be a mapping satisfying*

$$\psi(K \cap L) = \psi(K) \vee \psi(L), \tag{1}$$

$$\psi(K \vee L) = \psi(K) \cap \psi(L) \tag{2}$$

for all $K, L \in \mathcal{K}_{(0)}^d$. Then either ψ is constant, or there exists a linear transformation $g \in GL(d)$ such that $\psi(K) = gK^$ for all $K \in \mathcal{K}_{(0)}^d$.*

Geom. Funct. Anal 18 (2008)

Recap: On polarities and computational convexity

- Bregman divergences and convex duality
- Bregman divergences as power distances
- Conjugate functions: quadratic polarities & gauge freedom
- Neural computational convexity:
(Approximating untractable convex conjugates with neural networks)

Summary

- Bregman divergences are coordinatization of “point distance” on Hessian manifolds expressed in dual charts as dual Bregman divergences or in the mixed parameterization as a Fenchel-Young divergence or equivalently power distance in a domain of $\mathbb{R}^n$.
 - Computational Hessian information geometry from power distance CG
- Conjugate functions defined via quadratic polarity allows one to express the divergence as inner products (fast on GPU) and consider affine deformations
- When not available in closed form, conjugates can be approximated using neural networks where the gauge freedom is used to stabilize the learning/accuracy.
 - Neural information geometry = NN-approximations of IG

References

- Neural Legendre-Fenchel transform with Hessian Preconditioning, arXiv:2606.09077
- Minimum enclosing Bregman balls made easy, arXiv:2607.24197

Preconditioning Neural Legendre transform

paraboloid $Q(\theta) = \frac{1}{2}\|\theta\|^2$ is its own Legendre conjugate: $Q^* = Q$

with conjugate map **identity**: $\theta^*(\eta) = \eta$ (initialization in NN)

Taylor approximation:

$$f(\theta) \approx f(\theta_0) + \frac{1}{2}(\theta - \theta_0)^\top H(\theta - \theta_0), \quad H = \nabla^2 f(\theta_0).$$

Under the change of variable $z = H^{1/2}(\theta - \theta_0)$ this becomes $f \approx f(\theta_0) + \frac{1}{2}\|z\|^2$

learning task in **z-space** is approximately that of the canonical paraboloid (stable!)

$$(f')^{*'}(\eta') = H^{-1/2} \theta^*(H^{1/2}\eta') - \theta_0 =: z^*(\eta')$$

$$f'(z) = f(H^{-1/2}z + \theta_0) - f(\theta_0),$$

$$\eta' = \eta H^{-1/2} \quad (\text{row-vector convention}) \quad \theta^*(\eta) = H^{-1/2} z^*(\eta H^{-1/2}) + \theta_0$$

Convex duality via Legendre-Fenchel transform

- Legendre-Fenchel transform of a convex function F :

$$F^*(\eta) = \sup_{\theta \in \Theta} \{ \langle \theta, \eta \rangle - F(\theta) \} = \sup_{\theta \in \Theta} \{ H_{\theta}(\eta) \}$$

Supremum of affine functions. Involution $(F^*)^* = F$, **H/V rep of polytopes**

Graph(F): pointwise supremum of all affine functions majorized by F

- Convex conjugate: $F^*(\eta) = \langle (\nabla F)^{-1}(\eta), \eta \rangle - F((\nabla F)^{-1}(\eta))$
since supremum is obtained when $\nabla F(\theta) = \eta$.

- **Dual Bregman divergences**/swapping param.: $B_F(\theta_1 : \theta_2) = B_{F^*}(\eta_2 : \eta_1)$

Dual Fenchel-Young/Bregman divergences

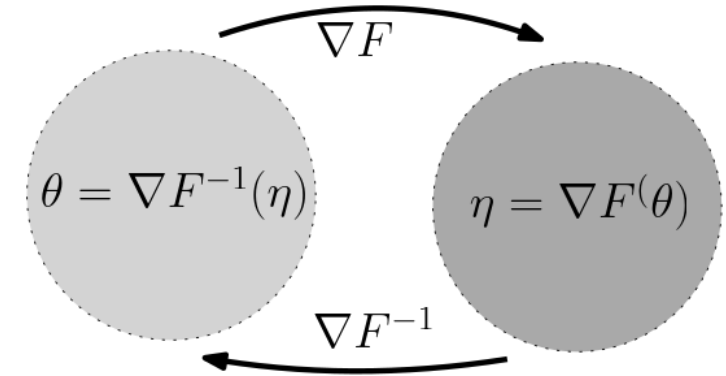
- Young inequality: $F(\theta_1) + F^*(\eta_2) \geq \langle \theta_1, \eta_2 \rangle$
with equality iff. $\eta_2 = \nabla F(\theta_1)$
- Build the **Fenchel-Young divergence** from this inequality: lhs-rhs ≥ 0
 $Y_{F, F^*}(\theta_1, \eta_2) = F(\theta_1) + F^*(\eta_2) - \langle \theta_1, \eta_2 \rangle \geq 0$
- Mixed and dual parameterizations of an underlying geometric divergence
 $\eta = \nabla F(\theta)$ and $\theta = \nabla F^*(\eta)$

$$B_F(\theta_1 : \theta_2) = Y_{F, F^*}(\theta_1 : \eta_2) = Y_{F^*, F}(\eta_2, \theta_1) = B_{F^*}(\eta_2 : \eta_1)$$

Dual Bregman/Fenchel-Young divergences

- **Gradient map** ∇F yields **dual parameterization**

$$\eta = \nabla F(\theta)$$



- **Dual Bregman divergences:** $B_F(\theta_1 : \theta_2) = B_{F^*}(\eta_2 : \eta_1)$

where F^* is the Legendre-Fenchel transform of F :

$$F^*(\eta) = \sup_{\theta \in \Theta} \{ \langle \theta, \eta \rangle - F(\theta) \}$$

$$\text{with } \theta = \nabla F^{-1}(\eta) = \nabla F^*(\eta)$$

- Build the **Fenchel-Young divergence** from Young inequality:

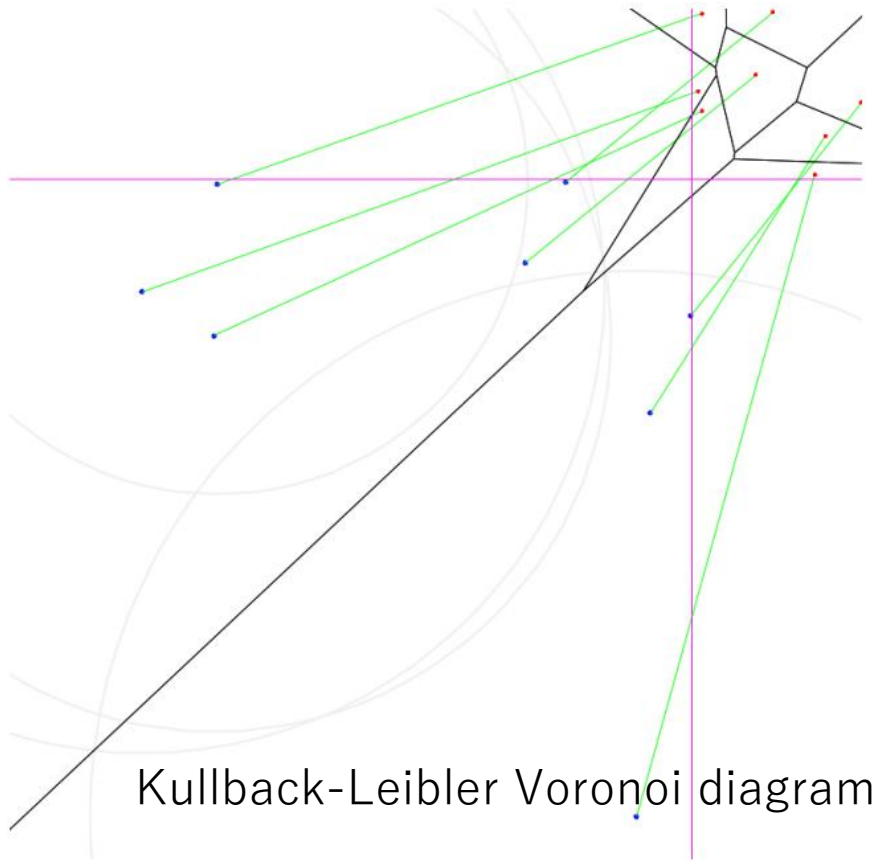
$$F(\theta_1) + F^*(\eta_2) \geq \langle \theta_1, \eta_2 \rangle$$

$$Y_{F, F^*}(\theta_1, \eta_2) = F(\theta_1) + F^*(\eta_2) - \langle \theta_1, \eta_2 \rangle \geq 0$$

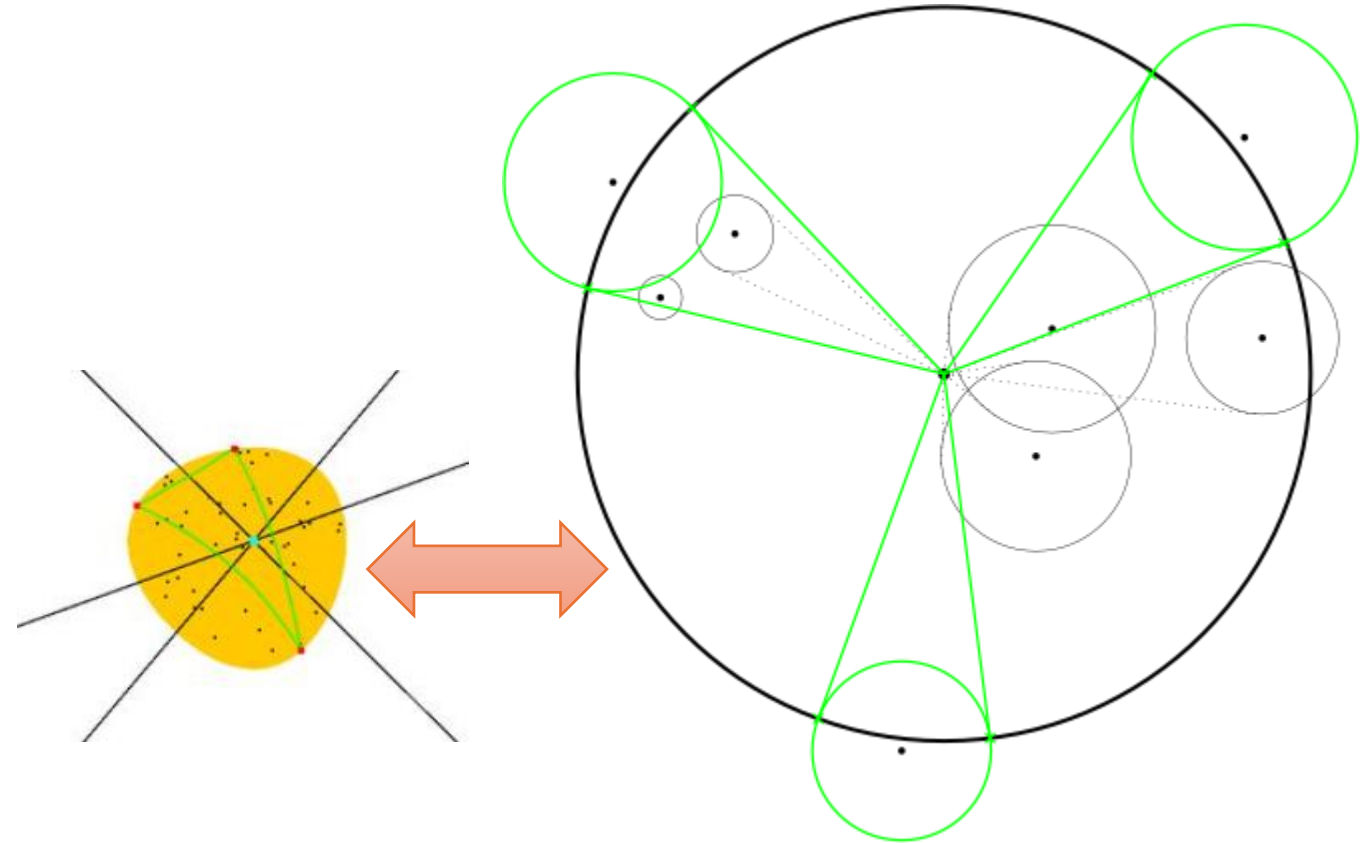
$$B_F(\theta_1 : \theta_2) = Y_{F, F^*}(\theta_1 : \eta_2) = Y_{F^*, F}(\eta_2, \theta_1) = B_{F^*}(\eta_2 : \eta_1)$$

Bregman divergences as power distances: CG

Use toolbox of computational geometry in information geometry!



Ex. 1: Bregman Voronoi in θ as
clipped power diagrams in η



Ex. 2: Bregman miniball in θ
As Power miniball in η